

Lab Exercise #2: Total Cost To Carpet 3 Rooms

1 Overview

In this lab, you will write a program to calculate the cost of carpeting 3 rooms. Carpet is priced by the square foot. The total cost includes the price of the carpet, labor and sales tax.

Your program should

- Request the user to input the carpet price per square foot (`double`), room width (`int`) and room length (`int`).
- Calculate and output the area of the room in square feet.
- Calculate and output the price of the carpet including an additional 20% for waste.
- Calculate and output the labor cost for installation (\$0.75 per actual square foot).
- Calculate and output the sales tax (4.5%) on both the carpet and labor cost.
- Calculate and output the total cost which includes the carpet price, labor and sales tax.
- Repeat the above for room 2 and room 3.
- Calculate and output the total cost for carpeting all three rooms.
- Separate and label the output for each order and Total Cost (ie, Order #1, Order #2 and Order #3).
- Any constants should be declared as `const` type within the body of your program NOT as directives.

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets `[]` in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #2 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #2 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below.

```
Mark Schwitzerlett and Suzy Q -- Lab #3

Please input the carpet price per square feet, room width, and room length for room 1 0.95 12 12
Order #1
Room: 144 sq ft
Carpet: $164.16
Labor: $108.00
Tax: $12.25
Cost: $284.41

Please input the carpet price per square feet, room width, and room length for room 2 1.25 8 18
Order #2
Room: 144 sq ft
Carpet: $216.00
Labor: $108.00
Tax: $14.58
Cost: $338.58

Please input the carpet price per square feet, room width, and room length for room 3 1.12 10 17
Order #3
Room: 170 sq ft
Carpet: $228.48
Labor: $127.50
Tax: $16.02
Cost: $372.00

Total Sales: $994.99

...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named `Lab#2_FirstName.LastName.c`

Due date: The lab is due at the end of your lab period, either Monday, February 9th or

Tuesday, February 10th.

Be sure to save a copy of your .c file.